IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATIONS 2024

WAVELETS AND APPLICATIONS SOLUTIONS

SOLUTIONS

1. (a)

$$\hat{X}(z) = \frac{G_0(z)}{2}\left[H_0(z)X(z) + H_0(-z)X(-z)\right] + \frac{G_1(z)}{2}\left[H_1(z)X(z) + H_1(-z)X(-z)\right].$$

Therefore PR condition is:

$$G_0(z)H_0(z) + G_1(z)H_1(z) = 2$$

and

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = 0.$$

(b) i. The term containing $X(-z)$ is

$$[G_0(z)H_0(-z) + G_1(z)H_1(-z)]X(-z).$$

If we set $G_0(z) = H_1(-z)$ and $G_1(z) = -H_0(-z)$ then the term containing $X(-z)$ becomes:

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = H_1(-z)H_0(-z) - H_0(-z)H_1(-z) = 0.$$

So the aliased component is removed.

ii. Since the aliased component is removed, the output is given by:

$$Y(z) = [G_0(z)H_0(z) + G_1(z)H_1(z)]X(z)$$

and using $G_0(z) = H_1(-z)$ and $G_1(z) = -H_0(-z)$ yields

$$Y(z) = [H_1(-z)H_0(z) - H_0(-z)H_1(z)]X(z).$$

If we assume $H_0(z) = (1+z^{-1})^2$ and $H_1(z) = -(1-z^{-1})^2(1+4z^{-1}+z^{-2})$ then

$$\begin{aligned} H_1(-z)H_0(z) &= -(1+z^{-1})^4(1-4z^{-1}+z^{-2}) \\ &= [(1+4z^{-1}+6z^{-2}+4z^{-3}+z^{-4})(-1+4z^{-1}-z^{-2}) \\ &= (-1+9z^{-2}+16z^{-3}+9z^{-4}-z^{-6}) = a \end{aligned}$$

and

$$\begin{aligned} -H_0(-z)H_1(z) &= (1-z^{-1})^4(1+4z^{-1}+z^{-2}) \\ &= [(1-4z^{-1}+6z^{-2}-4z^{-3}+z^{-4})(1+4z^{-1}z^{-2}) \\ &= (1-9z^{-2}+16z^{-3}-9z^{-4}+z^{-6}) = b \end{aligned}$$

Consequently

$$Y(z) = (a+b)X(z) = 32z^{-3}X(z).$$

This means that the system introduces a delay of 3 steps and amplify the signal by a factor 32.

A. If $x[n] = \delta[n]$ then $y[n] = 32\delta[n-3]$

B. if $x[n] = (-1)^n$ then $y[n] = 32 \cdot (-1)^{(n-3)}$

(c) i. $P(z) = (z^1 + 2 + z)/2$, consequently $P(z) + P(-z) = 2$.

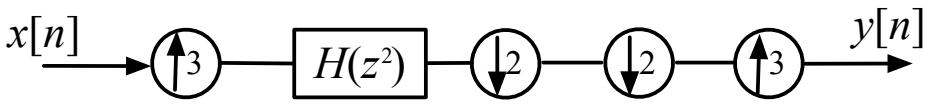
ii. Since this is an orthogonal filter bank then

$$G_1(z) = -z^{-1}G_0(-z^{-1}) = (1 - z^{-1})/\sqrt{2}$$

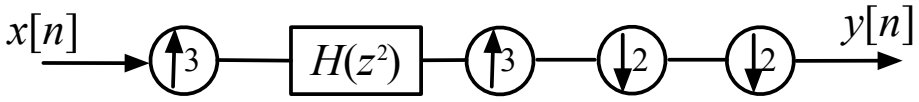
and

$$H_1(z) = G_1(z^{-1}) = (1 - z)/\sqrt{2}.$$

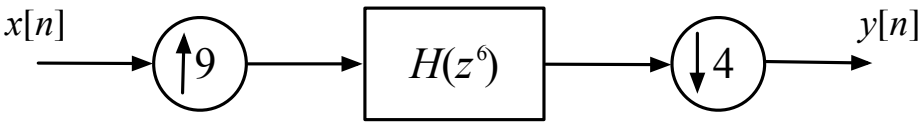
2. (a) We start by applying the first Noble identity and we obtain:



We then use the fact that the numbers three and two are co-prime to change the order of downsampling and upsampling leading to:

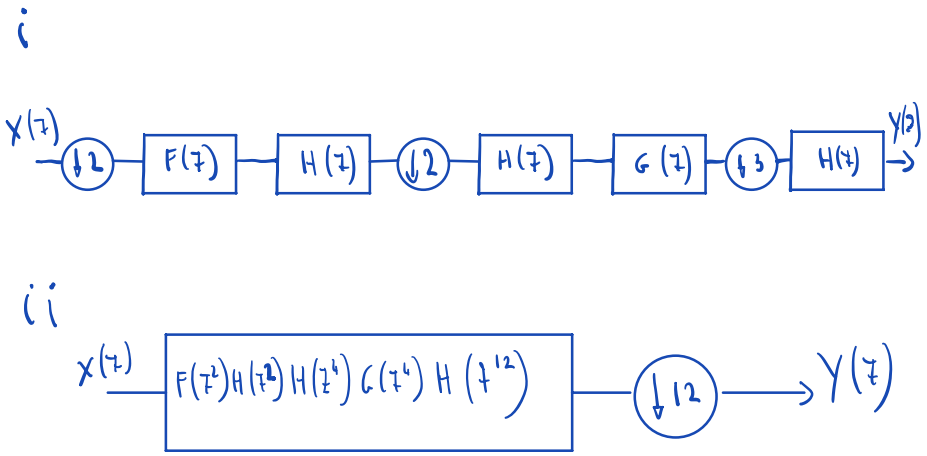


We finally apply the second Noble identity to obtain



Consequently, if $H(z) = z^{-2}$, then the equivalent filter $G(z)$ is given by $G(z) = H(z^6) = z^{-12}$.

(b) The equivalent representation is shown in the figure below:



(c) i. The input/output relationship is:

$$\hat{x}(z) = \frac{1}{2}[G(z^{1/2})H(z^{1/2}) + G(-z^{1/2})H(-z^{1/2})]X(z).$$

Consequently we require that $P(z) = G(z)H(z)$ satisfies

$$P(z) + P(-z) = 2.$$

ii.

$$G(z) = az^{-2} + bz^{-1} + c + bz + az^2.$$

Given $H(z)$, we need to find constants a, b, c such that $P(z) + P(-z) = 2$. We have that

$$\begin{aligned} P(z) &= H(z)G(z) \\ &= (az^{-3} + bz^{-2} + cz^{-1} + b + az + 2az^{-2} + 2bz^{-1} + 2cz + 2bz + az^2 \\ &+ az^{-1} + bz + cz + bz^2 + az^3)/4. \end{aligned}$$

The condition is then satisfied when $b + 2a = 0$ and $b + c = 2$. The solution is not unique but a possible option is $a = -1/4$, $b = 1/2$ and $c = 3/2$.

3. (a) i. We start with $\tilde{\varphi}_1(t)$. We want $\langle \tilde{\varphi}_1(t), \varphi_j(t) \rangle = \delta_{1,j}$ for $j = 1, 2, 3$. Since

$$\tilde{\varphi}_1(t) = \sum_{k=1}^{3} \alpha_{1,k} \varphi_k(t),$$

we have

$$\langle \tilde{\varphi}_1(t), \varphi_j(t) \rangle = \sum_{k=1}^{3} \alpha_{1,k} \langle \varphi_k(t), \varphi_j(t) \rangle = \delta_{1,j} \quad \text{for} \quad j = 1, 2, 3.$$

We also note that:

$$\langle \varphi_1(t), \varphi_1(t) \rangle = 2/3,$$
$$\langle \varphi_1(t), \varphi_2(t) \rangle = 1/6$$

and

$$\langle \varphi_1(t), \varphi_3(t) \rangle = 1/6.$$

Therefore, due to circulant symmetry, we have that

$$\langle \varphi_i(t), \varphi_j(t) \rangle = \begin{cases} 2/3, & i = j \\ 1/6, & i \neq j. \end{cases}$$

Consequently, the system of linear equations:

$$\langle \tilde{\varphi}_1(t), \varphi_j(t) \rangle = \sum_{k=1}^{3} \alpha_{1,k} \langle \varphi_k(t), \varphi_j(t) \rangle = \delta_{1,j} \quad \text{for} \quad j = 1, 2, 3$$

reduces to

$$\begin{cases} \frac{2\alpha_{1,1}}{3} + \frac{\alpha_{1,2}}{6} + \frac{\alpha_{1,3}}{6} &= 1 \\ \frac{\alpha_{1,1}}{6} + \frac{2\alpha_{1,2}}{3} + \frac{\alpha_{1,3}}{6} &= 0 \\ \frac{\alpha_{1,1}}{6} + \frac{\alpha_{1,2}}{6} + \frac{2\alpha_{1,3}}{3} &= 0 \end{cases}$$

with solution: $\alpha_{1,1} = 5/3$, $\alpha_{1,2} = -1/3$ and $\alpha_{1,3} = -1/3$. This implies that

$$\tilde{\varphi}_1(t) = \frac{5}{3}\varphi_1(t) - \frac{1}{3}\varphi_2(t) - \frac{1}{3}\varphi_3(t).$$

Moreover

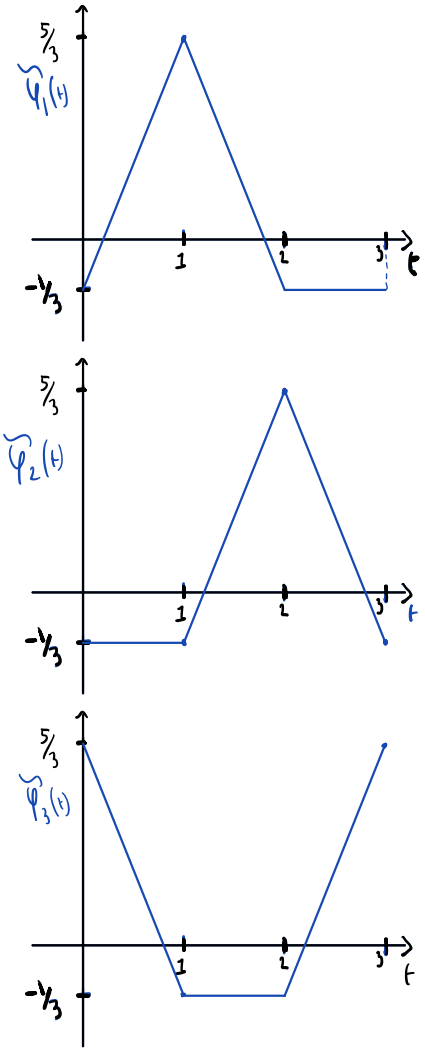
$$\tilde{\varphi}_2(t) = \tilde{\varphi}_1(t-1)$$

and

$$\tilde{\varphi}_3(t) = \tilde{\varphi}_1(t-2),$$

where the shifts are all circulant shifts.

ii. The sketch of the three functions is shown below:



(b) i. We have the following

$$\langle x(t), \tilde{\varphi}_1(t)\rangle = 2\int_0^1 (2t - 1/3)dt = 4/3,$$

$$\langle x(t), \tilde{\varphi}_2(t)\rangle = -\frac{1}{3}\int_0^1 dt + \int_0^1 (2t - 1/3)dt = 1/3,$$

and

$$\langle x(t), \tilde{\varphi}_3(t)\rangle = 1/3.$$

This means that, in this case,

$$x_v(t) = \sum_{i=1}^{3} \langle x(t), \tilde{\varphi}_i(t)\rangle \varphi_i(t) = \frac{4}{3}\varphi_1(t) + \frac{1}{3}\varphi_2(t) + \frac{1}{3}\varphi_3(t).$$

ii. We need to verify that: $\langle e(t), \varphi_1(t)\rangle = \langle e(t), \varphi_2(t)\rangle = \langle e(t), \varphi_3(t)\rangle = 0$. Indeed we have:

$$\begin{aligned}
\langle e(t), \varphi_1(t)\rangle &= \langle x(t), \varphi_1(t)\rangle - \langle x_v(t), \varphi_1(t)\rangle \\
&= 1 - \tfrac{4}{3}\langle \varphi_1(t), \varphi_1(t)\rangle - \tfrac{1}{3}\langle \varphi_2(t), \varphi_1(t)\rangle - \tfrac{1}{3}\langle \varphi_3(t), \varphi_1(t)\rangle \\
&= 1 - \tfrac{8}{9} - \tfrac{1}{18} - \tfrac{1}{18} \\
&= 0.
\end{aligned}$$

4. (a) We first note that $a[n]$ is the sampled version of $a(\tau) = \langle \varphi(t), \varphi(t-\tau) \rangle$ with sampling period $T = 1$. This means that if we denote with $\hat{a}(\omega)$ the Fourier transform of $a(\tau)$, then

$$A(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \hat{a}(\omega + 2\pi k), \tag{1}$$

where $A(e^{j\omega})$ is the discrete-time Fourier transform of $a[n]$. This is because sampling in time leads to a replication in frequency domain.
We are therefore just left with the problem of finding the exact expression for $\hat{a}(\omega)$. We have that:

$$\begin{aligned}
\hat{a}(\omega) &= \int_{-\infty}^{\infty} a(\tau) e^{-j\omega\tau} d\tau \\
&= \int_{-\infty}^{\infty} \langle \varphi(t), \varphi(t-\tau) \rangle e^{-j\omega\tau} d\tau \\
&= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \varphi(t)\varphi(t-\tau) dt \right] e^{-j\omega\tau} d\tau \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \varphi(t)\varphi(x) e^{j\omega x} e^{-j\omega t} dx dt \\
&= \int_{-\infty}^{\infty} \varphi(t) e^{-j\omega t} dt \int_{-\infty}^{\infty} \varphi(x) e^{j\omega x} dx \\
&= |\hat{\varphi}(\omega)|^2.
\end{aligned}$$

By replacing $\hat{a}(\omega) = |\hat{\varphi}(\omega)|^2$ in Eq. (1), we conclude the proof.

(b) We need to compute $a[n]$ for this case. By inspection we can see that $a[n] = 0$ for $n > 1$ and $n < -1$. Moreover,

$$a[0] = \langle \varphi(t), \varphi(t) \rangle = 2 \int_0^1 t^2 dt = 2/3$$

and

$$a[1] = a[-1] = \langle \varphi(t-1), \varphi(t) \rangle = \int_0^1 t(1-t) dt = 1/6.$$

Consequently

$$A(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \hat{a}(\omega + 2\pi k) = \frac{2}{3} + \frac{1}{3}\cos\omega$$

which implies that $A = 2/3 - 1/3 = 1/3$ and $B = 2/3 + 1/3 = 1$.

(c) i. $G_0(z) = \sqrt{2}\left(\frac{1+z^{-1}}{2}\right)^3$, therefore $G_0(e^{j\pi}) = 0$ and $G_0(1) = \sqrt{2}$, so the necessary conditions for convergence are satisfied.

ii. In this case, we can write $M_0(\omega)$ as follows

$$M_0(\omega) = \left(\frac{1 + e^{-j\omega}}{2}\right)^N R(\omega),$$

where $R(\omega) = 1$ and $N = 3$. This means that $B = \max_\omega R(\omega) = 1$ and since $B < 2^{N-2}$ we know that the sufficient condition for $\varphi(t)$ to be continuous with its first order derivative is satisfied. In fact, the iteration converges to the quadratic spline.

iii. $G_0(z)$ has three zeros at $\omega = \pi$, therefore the analysis wavelet has three vanishing moments.